LBP Face Recognition with Contrast Normalization and Spatially Tolerant Matching

Objective and protocol. Improve recognition under uneven illumination and rotation while preserving normal-image accuracy. ORL contains 40 subjects with 10 images each (92×112 pixels). Images 1–5 per subject form the gallery; images 6–10 form the 200-image test set. Only probe images are transformed. The original baseline uses uniform LBP with 16 neighbors, radius 2, nine regions (2187 features), and chi-square nearest-neighbor matching.

Conditions. Uniform brightness is $0.5I + 20$. Side lighting is $I[1 + s\,\mathrm{linspace}(-1, 1, W)]$, clipped to $[0, 255]$. Test rotations preserve image dimensions using bilinear interpolation and nearest boundary handling. All table accuracies are percentages.

Table 1. Original baseline

Condition	Accuracy (%)
Normal	97.5
Uniform brightness	97.5
Lighting $s = 0.2$	96.0
Lighting $s = 0.4$	94.5
Lighting $s = 0.6$	89.5
Lighting $s = 0.8$	70.5
Rotation 5°	97.0
Rotation 10°	92.5
Rotation 20°	63.0

Table 2. New method

Condition	Accuracy (%)
Normal	97.5
Uniform brightness	97.5
Lighting $s = 0.2$	95.5
Lighting $s = 0.4$	95.5
Lighting $s = 0.6$	95.5
Lighting $s = 0.8$	93.0
Rotation 5°	97.5
Rotation 10°	93.0
Rotation 20°	67.0

Improvement method. Local contrast normalization subtracts the Gaussian local mean and divides by the local standard deviation:

$$\widehat{I} = \frac{I - \mu}{\sqrt{\max(G_\sigma * (I^2) - \mu^2, 0) + 10^{-6}}}, \qquad \mu = G_\sigma * I.$$

Multiscale uniform LBP uses 16 neighbors at radii 1, 2 and 3. Regional histograms sum to one; chi-square costs are averaged across scales. Each region can match itself or an immediately adjacent region, with a Manhattan displacement penalty. The mean nearest-region costs in both directions are averaged. This is a spatially tolerant heuristic, not one-to-one assignment or full rotation invariance. Recognition uses no corrective image rotations and no logarithmic normalization.

Validation and distance combination. Images 1–4 per subject form the validation gallery; image 5 provides validation probes. Thirty candidates combine $\sigma \in \{5, 10\}$, penalties $\{0.02, 0.05, 0.10\}$, and weights $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$. Selection preserves validation normal/brightness accuracy, then maximizes mean accuracy over the other seven conditions. Selected values are $\sigma = 5$, penalty 0.02, and $\alpha = 0.5$. The final distance is $(1 - \alpha)D_{\text{base}}/(30 \cdot 37 \cdot 9) + \alpha D_{\text{spatial}}$. Parameters are fixed for the final 5/5 evaluation.

Table 3. Before and after comparison

Condition	Baseline (%)	New method (%)	Gain (pp)
Normal	97.5	97.5	0.0
Uniform brightness	97.5	97.5	0.0
Lighting $s = 0.2$	96.0	95.5	−0.5
Lighting $s = 0.4$	94.5	95.5	+1.0
Lighting $s = 0.6$	89.5	95.5	+6.0
Lighting $s = 0.8$	70.5	93.0	+22.5
Rotation 5°	97.0	97.5	+0.5
Rotation 10°	92.5	93.0	+0.5
Rotation 20°	63.0	67.0	+4.0

Conclusion and limitations. Normal accuracy is preserved at 97.5%; strong-lighting accuracy rises from 70.5% to 93.0%, and 20° rotation from 63.0% to 67.0%. Mild lighting ($s = 0.2$) decreases by 0.5 percentage points (pp). The three key workshop goals are met, but rotation improvement is modest. Multiscale region matching adds computation. Test results were viewed during earlier exploration, so this is not an untouched holdout; statistical significance and generalization to new data remain unverified.